

Name:

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CHEMICAL REACTION ENGINEERING

FINAL EXAM

CHE 321, SPRING 2025, CLOSED BOOK, TIME: 60 MIN

The endothermic gas phase reaction $A \rightarrow B + C$ is carried out in a single steam-jacketed fluidized bed reactor, where the temperature of the steam circulating through the jacket remains constant at $T_a = 300^\circ\text{C}$.

FBR

73.15K

For the following data, calculate the steady-state reactor temperature.

Conversion: $X = 85\%$ Heat-transfer area of the jacket: $A = 5 \text{ m}^2$ Overall heat transfer coefficient of the jacket: $U = 500 \frac{\text{J}}{\text{s} \cdot \text{m}^2 \cdot \text{K}}$ Heat of reaction per unit mole of species A at 25°C : $\Delta_r H = 120 \frac{\text{kJ}}{\text{mol}}$ Feed temperature: $T_0 = 40^\circ\text{C} = 313.15 \text{ K}$ Feed molar flow rates: $F_{A0} = 10 \frac{\text{mol}}{\text{s}}$, $F_{I0} = 2 \frac{\text{mol}}{\text{s}}$ (inert), $F_{B0} = F_{C0} = 0$ Heat capacities (assume constant) $c_{pA} = 90 \frac{\text{J}}{\text{mol} \cdot \text{K}}$, $c_{pB} = 60 \frac{\text{J}}{\text{mol} \cdot \text{K}}$, $c_{pC} = 40 \frac{\text{J}}{\text{mol} \cdot \text{K}}$, $c_{pI} = 50 \frac{\text{J}}{\text{mol} \cdot \text{K}}$ Reactor energy balance: $\frac{dU}{dt} = \sum F_{a0} h_a(T_0) - \sum F_a h_a(T) + \dot{Q} + \dot{W}_s$ Conversion $X \equiv \frac{F_{A0} - F_A}{F_{A0}}$

$$\begin{aligned} F_{A0} X &= F_{A0} - F_A \\ F_A &= F_{A0} - F_{A0} X \\ F_A &= F_{A0}(1-X) \end{aligned}$$

$$h = c_p T$$

$$\begin{aligned} \frac{dU}{dt} &= \sum F_{a0} h_a(T_0) - \sum F_a h_a(T) + \dot{Q} + \dot{W}_s \\ 0 &= (F_{A0} h_A(T_0) + F_{I0} h_I(T_0) + F_{B0} h_B(T_0) + F_{C0} h_C(T_0)) \\ &\quad - (F_A h_A(T) + F_I h_I(T) + F_B h_B(T) + F_C h_C(T)) + \dot{Q} \\ F_A &= F_{A0}(1-X) \quad F_{I0} = F_I \quad F_B = F_{B0} + F_{A0} X \quad F_C = F_{C0} + F_{A0} X \\ 0 &= (F_{A0} h_A(T_0) + F_{I0} h_I(T_0)) - (F_{A0}(1-X) h_A(T) + F_{I0} h_I(T) \\ &\quad + F_{A0} X h_B(T) + F_{A0} X h_C(T)) + \dot{Q} \end{aligned}$$

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everything is J/s
(energy flow rate)

$$0 = 10 \frac{\text{mol}}{\text{s}} \cdot \frac{90 \text{ J}}{\text{mol} \cdot \text{K}} \cdot 313.15 \text{ K} + 2 \frac{\text{mol}}{\text{s}} \cdot \frac{50 \text{ J}}{\text{mol} \cdot \text{K}} \cdot 313.15 \text{ K}$$

$$- [10(1-0.85) 90 T + 2(50) T + 10(0.85) T (60) + 10(0.85) (40) T]$$

$$- 500 \frac{\text{J}}{\text{s} \cdot \text{K}} (300 + 273.15 \text{ K}) (5 \text{ m}^2)$$

$$0 = 281835 \text{ J/s} + 31315 \text{ J/s}$$

$$- [135 T + 100 T + 51 T + 34 T]$$

$$- 1432875 \text{ J/s}$$

$$29822 = 135 T + 100 T + 51 T + 34 T$$

$$29822 = 320 T$$

$$T = 93.82 \text{ K}$$

Since the reaction is endothermic,
this answer does not make sense,
as the reaction needed heat to be
added (heated jacket vs. inlet temperature).
↳ outlet would not be this hot

$$T = 93.82 \text{ K}$$

← however is per mass
if $h = C_p \Delta T$

$$0 = 900 (T_o - T) + 100 (T_o - T)$$

$$- [320 (T_o - T)] - 1432875$$

$$0 = 281835 - 900 T + 31315 - 100 T$$

$$- [100208 - 320 T] - 1432875$$

$$= 313150 - 1000 T - 100208 + 320 T - 1432875$$

$$- 1219733 = -680 T$$

$$T = 1794 \text{ K}$$

that is also not correct

$$0 = 281835 + 31315$$

$$- [320 T] - 1432875 + 120 \cdot 100 \cdot 8.5 \text{ J/K}$$

↳ mol reacted
↓ ↓ needed to consider

this makes more

sense for
an endothermic
reaction (heat is being added)

$$0 = -99725 - 320 T$$

$$\text{final answer} \rightarrow T = 311.64 \text{ K}$$

